1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

Primary hypothesis: In NMIBC treated with intravesical BCG, TLS-like lymphoid aggregates do not uniformly predict a durable response when PD-L1 is upregulated (adaptive resistance) and when tumor mutational burden (bTMB) is low. Specifically, TLS-high tumors with PD-L1-low and high bTMB will exhibit higher rates of durable response than TLS-high tumors with PD-L1-high and/or low bTMB, and TLS-low tumors will have poorer responses irrespective of PD-L1/bTMB status. Mechanistically, PD-L1 upregulation within TLS-rich NMIBC dampens TLS-primed effector T cells, an effect amplified by low neoantigen burden (low bTMB). Conversely, TLS-high, PD-L1-low, and high bTMB tumors foster robust, durable anti-tumor T-cell activity and better BCG outcomes. Translationally, a triple biomarker phenotype—TLS-high + PD-L1-high + low bTMB—identifies patients least likely to benefit from BCG alone and who may require early addition of PD-1/PD-L1 blockade or alternative immunotherapeutic strategies.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	PD-L1, bTMB
Primary endpoint	therapy_response (BCG)

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PD-L1: PD-L1 correlated with bTMB) for the mechanistic link between adaptive checkpoint induction and tumor immunogenicity, and Core Path 3 (cross-cancer immunotherapy biomarker lessons) for translational rationale.

bTMB: bTMB/burden relationships provide a genomic basis for neoantigen availability; ADNP in CTCs and CTC-based assays offer a liquid-biopsy route to monitor dynamics; TP53/ER/HPV context can modify immune programs and explain heterogeneity.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	MSH6	Supporting modifier from knowledge graph

Secondary	ADNP	Supporting modifier from knowledge graph
Secondary	TP53	Supporting modifier from knowledge graph
Secondary	ER_status	Supporting modifier from knowledge graph
Secondary	HPV_status	Supporting modifier from knowledge graph
Secondary	CTCs	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: quantify TLS-like aggregates by histology/digital pathology,

Prediction 2: measure PD-L1 expression by IHC in tumor and immune compartments, and

Prediction 3: derive bTMB from a validated targeted sequencing panel. Outcomes include: therapy_response at 3–6 months (complete response), and durable response metrics (recurrence-free survival at 12 months).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- MSH6: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- ADNP: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.